Roll Number: _____ Group: _____

Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)

Department of Computer Science & Engineering
Session: 2025-26, 2nd Year (MST)

Note: Attempt all questions. Attempt all parts of one question at one place. Assume missing data if any.

BE- 2nd Year, 25-26 EVENSEM
Date: 14th March, 2026
Time: 02:00 Hours
Total marks: 40 (Weightage:30)

Course Code: UCS 520
Subject: Computer Networks
Name of Faculty: MCL, SKK, SHD, GVR, HMG, ASB, SUB, TG

Q.1.	(a) A host in Department A sends a large file to a server in Department B. The communication path is: Host → Switch → Router → Switch → Server. Answer the following with respect to this communication. i. With the help of a suitable layered diagram, Explain the process of encapsulation and decapsulation in the OSI model during communication between Department A and Department B. ii. At which layers of the OSI model do a switch and a router operate? For each device, identify the Protocol Data Unit (PDU) it processes. iii. State the behavior of the following addresses during transmission by indicating whether they change at each hop or remain unchanged end-to-end: MAC address and IP address.	(3+1+1)	CO1	L2
	(b) A university plans to design a new network infrastructure with the following requirements: All departments inside a single building must share printers, files, and applications. Multiple campuses located across the same city must be interconnected using high-speed optical fiber. A dedicated high-speed network must connect servers to centralized storage devices without affecting regular network traffic. Some devices like keyboards, mouse, and headsets must work wirelessly within short distances. Communication between administrative office and examination cell must allow simultaneous two-way communication. i. Identify and justify the appropriate network type for each requirement. ii. List and explain the five components of data communication involved in this system. iii. Identify the suitable data flow mode for the last requirement and explain why.	(2+2+1)	CO1	L1
Q.2.	(a) A video conferencing service operates between a university campus and a remote research center over a dedicated network link with a bandwidth of 50 Mbps. The distance between the two locations is 2500 km, and the propagation speed of the signal is 2×10^8 m/s. Each packet has a size of 1000 bytes, with a processing delay of 3 ms and an average queuing delay of 5 ms. During transmission, the packet arrival delays observed at the receiver are 60 ms, 68 ms, 65 ms, and 75 ms. Briefly explain how jitter can affect the quality of real-time applications such as video conferencing and VoIP. Calculate the following: i. Transmission delay per packet. ii. Propagation delay. iii. Total end-to-end latency per packet. iv. Throughput if 200 MB of video data is transmitted in 40 seconds.	(5)	CO1	L3
	(b) What are the different persistence schemes used in CSMA. Discuss the working of each scheme with suitable state-space diagram. Discuss the key limitations of CSMA/CD protocol in the context of wireless networks.	(3+2)	CO2	L2

Q.3.	(a) Draw the graph of the data stream 1011001110 using each of the following schemes, assuming that the last signal level has been positive. i. NRZ-L scheme ii. NRZ-I scheme iii. Polar RZ scheme iv. Differential Manchester scheme v. Pseudoternary scheme	(5)	CO1	L3
	(b) A sender is transmitting seven frames (sequentially numbered 0 through 6) to a receiver using a sliding window protocol. The sender's window size N is 7, allowing it to transmit all frames initially without waiting for acknowledgments. The transmission timeline is as follows: • The sender transmits frames 0, 1, 2, 3, 4, 5, and 6. • Frames 0, 1, 2, and 3 arrive at the receiver successfully. The receiver sends an acknowledgment (ACK) for these frames, which the sender receives, allowing its window to slide. • Frame 4 is transmitted but is lost in the network and never reaches the receiver. • Frames 5 and 6 arrive at the receiver successfully. However, because frame 4 is missing, they are received out of order. For the Selective Repeat protocol, assume the receiver is configured to send a Negative Acknowledgment (NACK) specifically for the lost frame. Answer the following questions for both Go-Back-N and Selective Repeat protocols. i. Draw suitable diagrams to illustrate the state of the sender/receiver windows for each protocol. ii. Assuming the sender's timer for frame 4 eventually expires, what range of frames will the sender have to retransmit? iii Considering the original transmissions and all necessary retransmissions until all seven frames (0-6) are successfully acknowledged by the receiver, calculate the total number of frames transmitted for each protocol.	(3+1+1)	CO2	L4
Q.4	(a) A DHCP server is configured for the network 181.14.16.0/24. The server IP address is 181.14.16.170, the DHCP address pool ranges from 181.14.16.180 to 181.14.16.220, and the lease time is 3600 seconds. Explain the complete DHCP message format that occurs when a new client requests an IP address from this server. Show the scenario of client and server interaction in the DHCP operation and also explain why the well-known ports 67 (server) and 68 (client) are used.	(2+3)	CO3	L2
	(b) A large number of consecutive IP address are available starting at 198.16.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.	(5)	CO3	L3

********** All the Best **********